

CYBERSECURITY CERTIFICATION ROADMAP

A beginner-to-advanced guide to choosing professional credentials

Purpose

Build a clear certification path from foundational knowledge to a professional specialisation, without wasting money on credentials you do not need.

THE ROADMAP AT A GLANCE

01	Foundation CCST Cybersecurity + networking basics
02	Core professional credential CompTIA Security+
03	Choose a direction Security operations, penetration testing, networking or cloud security
04	Advanced stage OSCP+, CISSP or CCSP when experience and career direction support it

Guiding principle: You do not need every certification in this guide. Choose a sensible sequence, practise the skills, and only pay for an exam when you are ready.

1. How to use this roadmap

Professional certifications are different from course-completion certificates. Most certifications in this guide require an exam, and some require periodic renewal or continuing education. Training may be free even when the certification exam is paid.

Recommended order

Start with fundamentals, learn networking, earn one core security credential, then specialise. Do not treat the list as a checklist where every item is mandatory.

Suggested foundation plan

Step	Focus	What you should be able to do
1	IT fundamentals	Explain operating systems, accounts, permissions, hardware and basic troubleshooting.
2	Networking	Understand IP addresses, DNS, DHCP, TCP/UDP, ports, switches, routers and subnetting basics.
3	Security basics	Explain common threats, access control, patching, backups, endpoint security and incident response.
4	Hands-on practice	Use a safe lab to practise system configuration, packet analysis and defensive troubleshooting.

Budget note

Exam prices change by country, taxes, voucher provider and date. Confirm the current official price and exam requirements before paying. Free training does not necessarily mean a free exam.

2. Stage 1 — Beginner foundation

Goal: gain enough basic knowledge to understand cybersecurity concepts and start preparing for professional exams.

1. Cisco CCST Cybersecurity

Level: Beginner

Entry-level, exam-based certification covering security fundamentals, common threats, endpoint protection, vulnerabilities and incident response.

Roadmap note: A reasonable first credential if you want a guided starting point. Cisco offers self-paced preparation material; verify the current exam fee and requirements.

[Official information](#)

2A. Cisco CCST Networking

Level: Beginner

Introduces network devices, connectivity, IP addressing and troubleshooting.

Roadmap note: A good beginner networking option. Choose this if you want a structured entry-level path.

[Official information](#)

2B. CompTIA Network+

Level: Beginner to intermediate

Vendor-neutral networking certification covering network concepts, implementation, operations, troubleshooting and security basics.

Roadmap note: Alternative to CCST Networking. You generally do not need both; compare exam costs, study style and job requirements.

[Official information](#)

Decision: choose one networking route

For a lower-cost learning path, study networking basics for free first. If you later want a networking-focused professional credential, consider CCNA. Network+ is the vendor-neutral alternative. You do not need CCST Networking, Network+ and CCNA all before Security+.

3. Stage 2 — Core professional certifications

Goal: develop a broad security foundation that can support several entry-level cybersecurity career paths.

3. CompTIA Security+

Level: Intermediate

Covers threats and vulnerabilities, security architecture, identity and access management, security operations, incident response and risk management.

Roadmap note: Recommended main milestone for a broad, vendor-neutral cybersecurity certification. Build networking knowledge before sitting the exam.

[Official information](#)

Alternative. Cisco CCNA

Level: Intermediate

Focuses on networking, including network access, IP connectivity, IP services, security fundamentals and automation.

Roadmap note: Choose CCNA instead of Network+ if you prefer Cisco-oriented networking and want deeper routing and switching skills. It is not a dedicated cybersecurity certification.

[Official information](#)

Optional. Cisco Cybersecurity Associate (CBROPS)

Level: Intermediate

Centres on security operations, monitoring, host and network analysis, and incident investigation.

Roadmap note: Consider it if your target is a SOC or defensive security role. Check the current exam name, syllabus and fee on Cisco's official site.

[Official information](#)

Core recommendation

For a broad cybersecurity path, aim for Security+ after foundational study. If you discover that networking is your strongest interest, prioritise CCNA. Do not pay for several overlapping credentials before you have a clear reason.

4. Stage 3 — Choose a specialisation

After foundational security study, choose a branch based on the type of work you want to do. These options are alternatives, not a sequence that everyone must complete.

Blue team. CompTIA CySA+

Level: Intermediate to advanced

Focuses on security monitoring, threat detection, vulnerability management, analysis and incident response.

Roadmap note: Consider for SOC analyst and defensive security work after you understand networking and core security concepts.

[Official information](#)

Red team. CompTIA PenTest+

Level: Intermediate to advanced

Covers penetration-testing methods, vulnerability assessment, validation and reporting.

Roadmap note: Choose if you enjoy authorised security testing. Practise only in systems and labs where you have permission.

[Official information](#)

Cloud. Cloud security credentials

Level: Varies by provider

Cloud security paths may cover identity, access, logging, data protection, network controls and cloud architecture.

Roadmap note: Learn cloud administration first. Consider a relevant AWS or Microsoft credential only after choosing the cloud platform and role.

[Official information](#)

How to choose

If you enjoy...	Explore...	Example next step
Monitoring alerts and investigating incidents	Blue team / SOC	CySA+ or Cisco Cybersecurity Associate
Finding weaknesses in systems	Red team / penetration testing	PenTest+ then practical labs
Cloud platforms and identity controls	Cloud security	Cloud administration, then provider-specific security
Routers, switches and network design	Networking / network security	CCNA, then security specialisation

5. Stage 4 — Advanced credentials

These are longer-term goals. They can be expensive or require substantial experience; do not treat them as beginner certifications.

8. OSCP+

Level: Advanced practical penetration testing

A demanding hands-on credential focused on practical penetration testing and exploitation in an exam environment.

Roadmap note: Pursue only after building strong Linux, networking, scripting and lab skills. Check the current exam format and package pricing.

[Official information](#)

9. ISC2 CISSP

Level: Advanced security professional

Broad coverage of security and risk management, asset security, architecture, network security, identity, operations and software security.

Roadmap note: Full certification generally requires five years of qualifying work experience across at least two domains, subject to applicable waiver rules. Candidates who pass but lack the required experience may qualify as Associates of ISC2.

[Official information](#)

10. ISC2 CCSP

Level: Advanced cloud security

Focuses on cloud security architecture, design, operations, compliance and risk.

Roadmap note: Best suited to professionals with cloud and security experience. Check the current experience requirements before planning for the exam.

[Official information](#)

Important

Advanced certifications are valuable when they match your work and experience. A beginner is usually better served by strong fundamentals, one core certification and hands-on practice than by rushing toward advanced exams.

6. A practical path to follow

Phase	Action	Outcome
1	Study IT basics and networking; prepare for CCST Cybersecurity.	Understand basic security and earn a first credential.
2	Choose CCST Networking or Network+; practise IP addressing and troubleshooting.	Build networking foundations.
3	Prepare for CompTIA Security+.	Earn a broad security credential.
4	Choose blue team, red team, cloud or networking specialisation.	Develop role-specific skills.
5	Build work experience and consider advanced credentials when eligible.	Progress toward specialised or senior roles.

Free or official learning resources

- [Cisco Networking Academy](#) — Networking and cybersecurity learning material.
- [Microsoft Learn](#) — Windows, identity, security and cloud learning paths.
- [CompTIA](#) — Official certification objectives and exam information.
- [ISC2](#) — Official requirements for CISSP, CCSP and other credentials.
- [OffSec](#) — Practical penetration-testing training and certification information.

Before you pay for an exam

- Confirm the current exam code, syllabus, price, taxes and test availability on the provider's official website.
- Check whether the credential requires renewal or continuing education.
- Review job postings for your target role and note which certifications employers actually request.
- Use authorised practice labs; never scan or test an employer's network without explicit permission.
- Certifications support career growth but do not guarantee hiring, promotion or a specific salary.

Recommended milestone: build strong networking fundamentals, earn one beginner credential, and then target CompTIA Security+ as your main broad cybersecurity certification.